

THE ACTIVATION GRAPH AT WIDTH 1024: VERIFICATION, THE WALL-CHAIN GAP LAW, AND THE QUANTITIES THE CERTIFIED-APPROXIMATION THEOREM NEEDS

STAGE 24 OF THE QUANTUM-CUT / NONCOMMUTATIVE-GEOMETRY PROGRAMME

ABSTRACT. This note is written against the companion *Weighted activation graphs and certified approximation* (10 October 2026). It does three things. (i) Every finite claim of the companion was re-derived and checked independently (ten checks, exact rationals where finite; all pass). Its wall formula $\mu = \sum_e S_e j_e$ is re-proved in a Stein form, $\mathbb{E}F = \mathbb{E}\Delta F$, which shows that the layerwise Gaussian closure is *exactly* the mean-field version of the wall sum. (ii) For the physical wall graph we prove the one-particle upper bound $\text{gap} \leq 1/\lambda_{\max}(\Gamma^{-1/2} \text{Cov}(g) \Gamma^{-1/2})$, where g is the gate vector and Γ the diagonal of one-way wall fluxes. Numerically the bound is attained to within 0.5% on every Gaussian sign law tested, and exactly on the companion’s own crowded-sector family. Its obstruction in §6.1 is therefore a statement about the top eigenvalue of the gate covariance. (iii) At width 1024 on the official networks we measure the quantities that the companion’s remaining theorem depends on. These are the active-cardinality and excitation levels, the wall rates, the spectral-independence radius, cross-depth gate dependence, the wall-chain gaps, and the visible spectrum of the terminal ellipsoid with the location of first-pass errors in it.

1. FOUR INDICES, MEASURED

The companion separates MLP depth ℓ , active cardinality $r = |S_\ell|$, information level k and spectral depth m . Two of these can be read off the real networks directly (`scripts/s24_kikuchi.py`, net 0, $N = 2^{16}$ and 2^{17} Gaussian inputs, Table 1).

Active cardinality is pinned. On every layer of net 0, $r/n = 0.50 \pm 0.03$ in the mean. Its standard deviation is $0.0155n$ at layer 1, the binomial value $\sqrt{n}/2$ for independent fair gates, and falls to $0.008n$, about 8 neurons, at depth. That is tighter than independent coins. The level is also decoupled from the collective mode, because the gates of neurons with $a_i = w_i \cdot \hat{\mu} > 0$ and $a_i < 0$ respond to it in opposite directions. Consecutive levels are uncorrelated ($|\text{Corr}(r_\ell, r_{\ell+1})| \leq 0.06$). The next level is predicted twice as well by the current activation *values* as by the current *signs*: linear R^2 is 0.44–0.60 against 0.26–0.37. This quantifies §1.1 of the companion at width 1024.

The excitation level shrinks with depth. Let $S_\ell^{\text{ref}} = \{i : \alpha_{\ell,i} > 0\}$ be the closure’s reference pattern. The number of gates in which an input deviates from it, $|S_\ell \Delta S_\ell^{\text{ref}}|$, falls from $n/2$ at layer 1 to $0.31n, 0.25n, \dots, 0.085n$ (about 87 gates) at layer 16. The closure’s prediction $\sum_i \Phi(-|\alpha_{\ell,i}|)$ agrees to three digits, and the spread is 10–16 gates. A partial assignment that records the deviations from S^{ref} is therefore much shorter at depth than one recording the active set.

2. THE WALL GRAPH: THREE INDEPENDENT DERIVATIONS

2.1. The wall formula in Stein form. Let $F = a_L$, which is piecewise linear and positively homogeneous with $\nabla F = M(x) = D_L W_L \cdots D_1 W_1$. Euler’s identity and Stein’s lemma, applied twice, give $\mathbb{E}F(X) = \mathbb{E}[M(X)X] = \mathbb{E}\Delta F$. In generic position the distributional derivative of a

gate is $\partial_j \mathbf{1}\{z_{\ell,i} > 0\} = \delta(z_{\ell,i}) \partial_j z_{\ell,i}$, and $e_i^\top W_\ell D_{\ell-1} \cdots W_1 = \nabla z_{\ell,i}^\top$ on the wall. Hence

$$(1) \quad \mu = \sum_{\ell=1}^L \sum_{i=1}^n \mathbb{E} \left[\delta(z_{\ell,i}) \|\nabla z_{\ell,i}\|^2 A_\ell(X) e_i \right] = \sum_{\ell,i} \int_{\{z_{\ell,i}=0\}} \|\nabla z_{\ell,i}\| A_\ell e_i d\sigma_\gamma, \quad A_\ell = D_L W_L \cdots D_{\ell+1} W_{\ell+1}.$$

By the co-area formula this is the companion's $\sum_e S_e j_e$ with $j_e = \|v_e\| A_{>b} e_i$, its eq. (12).

Remark 2.1 (Generic position is needed for the δ form). On narrow nets ($n = 6, 8, L = 3, 4$) whole hidden layers switch off with positive probability. A pre-activation can then vanish identically on an open set, and the δ form of (1) gives wrong values. On those nets a window-estimator check misses by 30%. The aggregate-jump form, the companion's eq. (9), remains valid there. On generic nets ($n = 24$ with $L = 3$, and $n = 32$ with $L = 4$, with no dead hidden layer in 2×10^6 samples) the window and flip-count estimators of (1) both agree with direct Monte Carlo to within their sampling error, 2–5% max over neurons (`theory24/check_kink.sum.py`).

2.2. The rotation clock. Under the Mehler rotation $X_s = X \cos s + Y \sin s$ (Y an independent copy), paths are smooth and the gate process jumps along single-flip arrows. The velocity $\dot{z} = \nabla z \cdot Y$ is independent of X at $s = 0$, so Rice's formula gives the stationary one-way flux through a wall e :

$$(2) \quad c_e = \frac{1}{2} \mathbb{E}[\|\dot{z}\| \delta(z) \mathbf{1}_e] = \frac{1}{2} \sqrt{2/\pi} S_e = S_e / \sqrt{2\pi} \quad \text{per unit angle.}$$

At layer 1 this is $c = (2\pi)^{-1} \Pr(\text{signs of } z_{-i} \mid z_i = 0)$, a conditional orthant probability with the Schur-complement covariance. Its flip rate is exactly $1/\pi$ per neuron, whatever W_1 is. The correlated pair (X, X_s) has the law of the companion's OU pair when $\cos s = e^{-t}$. Then $s \approx \sqrt{2t}$, and (2) reproduces its constant: $\Pr(X \in C, X_t \in D) \approx \sqrt{2t} S_{CD} / \sqrt{2\pi} = \sqrt{t/\pi} S_{CD}$. Both were checked numerically: facet fluxes against Schur-complement orthants, and the $\sqrt{t/\pi}$ constant.

2.3. The Gaussian closure is the mean-field wall sum.

Proposition 2.2. *Let $m_\ell^{\text{cl}} = \sigma_\ell \varphi(\alpha_\ell) + \Phi(\alpha_\ell) \circ (W_\ell m_{\ell-1}^{\text{cl}})$ be the closure recursion, and $T_{L \leftarrow \ell} = \prod_{k>\ell} \text{diag } \Phi(\alpha_k) W_k$ its tangent. Then exactly*

$$m_L^{\text{cl}} = \sum_{\ell=1}^L T_{L \leftarrow \ell} (\sigma_\ell \circ \varphi(\alpha_\ell)).$$

Proof. Unroll the recursion; $m_0 = 0$. □

Compare (1). The closure replaces the wall measure-weighted gradient, $\mathbb{E}[\delta(z_{\ell,i}) \|\nabla z_{\ell,i}\|^2]$, by its Gaussian value $\sigma_{\ell,i} \varphi(\alpha_{\ell,i})$. It also replaces the conditional downstream jump $\mathbb{E}[A_\ell e_i \mid \text{wall}]$ by the product of averaged gates $T_{L \leftarrow \ell} e_i$. Every closure error is therefore exactly a sum over layers of a *wall-measure* error and a *downstream-jump* error. The identity holds to 10^{-16} (`theory24/check_kink.sum.py`). At width 1024 the aggregate wall rate per layer is predicted by the Gaussian Rice formula $(1/\pi) e^{-\alpha_i^2/2} \sqrt{\mathbb{E} \dot{z}_i^2} / \sigma_i$ to within 0.1–0.6% on every layer (Table 1). The wall-measure channel is thus not where a first pass loses accuracy in aggregate.

3. THE GAP OF THE PHYSICAL WALL CHAIN

Fix one layer. Let $g \in \{0, 1\}^n$ be its gate vector with law p . Its wall chain has one-way conductances $c_i(s_{-i})$ for flipping gate i at the rest-pattern s_{-i} , and $\Gamma_i = \sum_{s_{-i}} c_i(s_{-i})$ is gate i 's total one-way flux, half its flip rate. For Gaussian z_i with standardised mean α_i , $\Gamma_i = e^{-\alpha_i^2/2} / (2\pi)$.

Proposition 3.1 (One-particle bound). $\text{gap} \leq \lambda_{\text{lin}} := \frac{1}{\lambda_{\text{max}}(\Gamma^{-1/2} \text{Cov}_p(g) \Gamma^{-1/2})}$. At $\alpha = 0$ this is $\lambda_{\text{lin}} = \frac{2/\pi}{\lambda_{\text{max}}(\text{Corr}(g))}$.

Proof. Test $f = v \cdot g$. Every arrow of gate i changes f by v_i . Hence the Dirichlet form is $\mathcal{E}(f, f) = \sum_i v_i^2 \Gamma_i$, while $\text{Var } f = v^\top \text{Cov}(g) v$. Minimise the quotient. $\square$

Proposition 3.2 (Two gates). *For two gates of a centred Gaussian pair with correlation ρ , the gap equals $\lambda_{\text{lin}} = (2/\pi)/(1 + \frac{2}{\pi} \arcsin \rho)$. For two unit rows at angle $\pi\varepsilon$ this is $1/(\pi(1 - \varepsilon)) \geq 1/\pi$.*

Proof. The four patterns form a cycle with stationary law (a, b, a, b) , $a = \frac{1}{4} + \frac{\arcsin \rho}{2\pi}$. Given $z_i = 0$ the other coordinate is symmetric, so all four conductances equal $1/(4\pi)$. The reflections (swap the two gates; flip both) diagonalise the generator. $g_1 + g_2 - 1$ has Rayleigh quotient $1/(2\pi a)$, $g_1 - g_2$ has $1/(2\pi b)$, and the centred parity has at least $4/\pi$. Also $\text{Corr}(g_1, g_2) = \frac{2}{\pi} \arcsin \rho$ by Sheppard's formula. $\square$

The contrast with the information walk is the point of Proposition 3.2. For the same two rows the companion's random-site heat-bath walk has spectrum $\{1, 1 - \varepsilon, \varepsilon, 0\}$, so its gap $\varepsilon \rightarrow 0$ as the rows align. The physical chain instead passes quickly through the thin wedge cells. Parallel walls are not an obstruction for it.

Remark 3.3 (Near-equality, and the companion's crowded sectors). Exact enumeration (`theory24/gap_level1.py`, Genz orthants at relative tolerance 10^{-6}) gives $\text{gap}/\lambda_{\text{lin}} \in [0.995, 1.000]$ for every Gaussian sign law tested. The cases cover $n = 5, 7, 8$; square Gaussian rows; two collective modes plus noise; ill-conditioned Wishart covariances; and standardised means up to $N(0, 2^2)$. The gap eigenvector is 99.7–99.995% linear in the gates. Equality is *not* exact once means are present (ratios 0.9948–0.9997), so the statement is a near-equality, not an identity.

For the companion's §6.1 family (n equiangular lines in $\mathbb{R}^2$, $m = 2n$ sectors) the ratio is 1.0000, and $\lambda_{\text{max}}(\text{Corr } g) = 4n/\pi^2$ exactly as n grows. Its local gap $\Theta(1/m)$ is therefore precisely $(2/\pi)/\lambda_{\text{max}}(\text{Corr } g)$. The obstruction is a large top eigenvalue of the gate covariance, produced by many nearly parallel walls in few dimensions.

Conjecture 3.4. *For the wall chain of any Gaussian sign law, $\text{gap} \geq c \lambda_{\text{lin}}$ with an absolute constant c ; numerically $c \geq 0.99$.*

Remark 3.5 (Isoperimetry does not suffice). The Gaussian isoperimetric inequality gives every wall chain the conductance $h \geq 1/\pi$ (sets of mass at most $\frac{1}{2}$; use $I(p)/p \geq 2\varphi(0)$). Cheeger's inequality, however, divides by the maximal out-rate, which grows without bound on thin cells. The companion's crowded-sector family shows that the gap really does vanish, so a gap theorem must use the gate covariance, as Proposition 3.1 does.

Random rows in high dimension are not crowded. For square random W the row correlations follow the Marchenko–Pastur law with top eigenvalue 4. Sheppard's formula then gives $\text{Corr}(g) \approx (1 - \frac{2}{\pi})I + \frac{2}{\pi}R$, so $\lambda_{\text{max}}(\text{Corr } g) \approx 1 + 3 \cdot \frac{2}{\pi} = 2.91$. The measured layer-1 value on the official nets is 2.95, and $\lambda_{\text{lin}} \approx 0.21$ per unit rotation angle. Deeper layers have larger gate correlations, but their wall rates are also more heterogeneous: gates that are nearly determined have both small variance and small flux. In the rate-normalised form of Proposition 3.1 the bound stays at 0.20–0.33 on every layer (Table 2). For deep layers the gate law is not Gaussian; there Proposition 3.1 is still an upper bound, and the near-equality is conjectural.

4. DEPENDENCE OF THE GATE LAW AT WIDTH 1024

Section 5 of the companion composes a Gaussian observation tier with a conditional lower tier of gap δ . Its §2.2 explains that spectral independence would supply δ , provided influence is controlled under every pinning. The unpinned influence radius $\eta = \lambda_{\text{max}}(\text{Corr } g) - 1$ is a necessary condition for that, and it is measurable (Table 2).

- η is about 2 at layer 1, matching the Marchenko–Pastur prediction $3 \cdot 2/\pi = 1.91$. It rises to 6–8 in the deep layers. The gate correlation spectrum is broad, with participation ratio 520–720 out of 1024. Even unpinned, the law is far from the $\eta < 1$ regime.

- Conditioning on k principal components of the pre-activations lowers η only slowly: at $k = 16$ it is still 4–5 at depth, and at $k = 128$ it is 1.2–1.6. The Marchenko–Pastur noise edge at these sample sizes is 0.18–0.27.
- Gates do not decorrelate across depth. $\|\text{Corr}(g_\ell, g_{\ell+d})\|_{\text{op}}$ is 2.4 at layer 1 and 6–7 at depth, with essentially no decay over $d = 1, 2, 4$.
- The closure’s level-1 depth gain $\prod_{k>\ell} g_k$, with $g_k = 2\overline{\Phi(\alpha_k)^2}$, is 0.03 from layer 1, 0.31 from layer 8 and 0.6–0.9 over the last four layers. Unresolved late-layer structure is not forgotten before the output.

The wall-chain bound of Proposition 3.1 is reported per layer from the measured gate covariance and wall rates. It is 0.20–0.33 per unit rotation angle on every layer, against $2/\pi = 0.64$ for independent gates. Removing 16 principal components raises it to 0.23–0.65; removing 128 raises it to 0.33–1.7. Gates with large $|\alpha|$ have both small variance and slow walls, which offsets the larger η at depth. The physical single-layer chain is therefore never degenerate on these networks, while the information heat-bath walk has no comparable guarantee.

The level process relaxes at rate 0.65 at layer 1, the independent-gate value $2/\pi$, and at 2–3 at depth. That stiffness is what pins r at $n/2$. Whether the joint chain over all nL gates, the companion’s full-history graph, keeps this rate is a separate question, because gate dependence does not decay across depth. Section 5 measures it.

5. THE FULL-HISTORY WALL GRAPH

The companion’s geometric graph uses complete gate histories. Proposition 3.1 applies to any set of gates, so it bounds that graph by the rate-normalised covariance of all nL gates at once (`scripts/s24_fullgates.py`). On net 0 there are 14,748 live gates in $N = 2^{16}$ inputs.

- The stacked gate correlation has many large eigenvalues: 71, 63, 59, 56, 55, 53 against a noise edge of 2.2. The rate-normalised covariance has top eigenvalues 41.6, 37.8, 35.9, 34.1, 32.9, 32.4. Hence

$$\text{gap}(\text{full-history wall chain}) \leq 1/41.6 = 0.024 \text{ per unit rotation angle,}$$

about nine times below any single layer.

- The bound falls steadily with the number of layers taken together. The best windows give 0.21 for one layer, 0.12–0.17 for two, 0.064–0.089 for four, 0.038–0.043 for eight and 0.024 for all sixteen.
- The slowest mode is spread over the whole depth. Each layer holds 3.5–8.7% of the top eigenvector, with most in layers 5–12.

Cross-depth coherence is what makes the local full-history chain slow, not any single layer. In the companion’s terms this is the quantitative content of “local adjacency does not inherit the dense gap” for the real networks. The actual rotation process decorrelates within an angle of $\pi/2$, while its local Markov shadow needs about $1/0.024 \approx 40$. A two-tier construction must therefore place these coherent cross-depth modes in its global observation tier, and the count of such modes sets the dimension of that tier. The count of such modes was not measured; that run was stopped in favour of theory (Stage 25).

6. THE TERMINAL ELLIPSOID AT WIDTH 1024

For hidden states equal to inputs, the companion’s operator $\mathcal{B}$ in eq. (44) has the nonzero spectrum of S/n , where $S = \text{Cov } F(X)$ is the output covariance. A first-pass error $e = \mu - b$ can be resolved in the eigenbasis of S (`scripts/s24_ellipsoid.py`; net 0 also from the larger run behind `scripts/s24_terminal.py`). Table 3 gives the results.

- The visible spectrum is concentrated. The participation ratio of S is about 51. The top 16, 64 and 256 eigenvectors carry 42%, 75% and 95% of the trace, and the bottom half carries 1%.

- First-pass errors live where the variance lives. For the closure, 69% of $|e|^2$ is in the top 16 eigenvectors, which is the collective tier. CC1 removes most of that, and its remaining error is still 86% inside the top 256. The v56 error is the most spread, with 6% in the bottom half. The Rayleigh ratio $e^\top S e / (|e|^2 \bar{\lambda})$ is 66, 8.3 and 5.1 for closure, CC1 and v56 on net 0.
- Sampling at the floor budget ($M \approx 4500$ samples after a CC1-sized pass) resolves this residual only with information about its direction. We give three oracle bounds for CC1 on net 0, each assuming e is known.

Orthogonal projections. The best $\mu = b + P(\bar{Y} - b)$ has rank at most one, because $ee^\top - S/M$ has at most one positive eigenvalue. Its risk is 1.3×10^{-7} , and the projection onto $e/|e|$ gives 1.4×10^{-7} . Both are about four times the floor target 3.1×10^{-8} .

Per-direction choice. Keeping or dropping each eigendirection with oracle knowledge gains 2% over CC1.

Oblique filter. The best linear filter estimates one coefficient along e through the whitened statistic $e^\top S^+(\bar{Y} - b)$. Its risk is $|e|^2 / (1 + M e^\top S^+ e) / n = 8.0 \times 10^{-9}$, below the target, because $M e^\top S^+ e = 233$. That signal-to-noise comes from the error's small components along the *quiet* directions, pooled coherently.

Hence a response direction predicted accurately where S is small would make one Mahalanobis-weighted calibration sufficient. None of the weight-derived families tried (coherent functions of a , propagated coherent functions, tangent singular subspaces, readout shapes, the CC1–closure direction) captures more than 86% of $|e|^2$ even orthogonally. They are built in the loud directions.

Variance reduction does not change this. Exact-mean input controls of degree $\leq k$ reduce variance by at most 1.23, 1.52, 2.10 and 3.38 for $k = 1, 2, 4, 8$. These are the chaos shares of the depth-16 arc-cosine kernel (`theory24/chaos_spectrum.py`); the measured linear control gives 1.23. No sampling density helps by more than $(\mathbb{E}|z|)^2 / \mathbb{E}|z|^2 \approx 1\%$ for the full output, because $|z|$ concentrates.

The residual that matters in (44) therefore sits mostly in the same ~ 50 –250 modes that carry the output variance, with a small but statistically decisive part in the quiet directions. The companion's deterministic route has no sampling variance: one occupation row and operator actions replace Monte Carlo. Its open obligation is a representation of the operator action that is accurate on the loud modes, with the quiet-direction part as a natural target for a final whitened calibration.

7. WHAT THE THEORY AND THE MEASUREMENTS NOW SAY TO BUILD

- (1) **The operator.** Use the physical wall chain, or a block or collective refinement of it, not single-gate heat-bath resampling. Its gap is governed by the rate-normalised gate covariance (Proposition 3.1, Remark 3.3), and it is bounded away from zero on every single layer of these networks: the level-1 values are 0.20–0.33 per unit rotation angle. The full-history graph is the open case (Section 5).
- (2) **The slow block.** About 100–250 modes are needed, by three independent measures. These are the visible spectrum of $\mathcal{B}$ (95% of trace in 256 modes), the location of CC1's error (86% in 256 modes), and the conditioning that brings the gate law near $\eta \approx 1$ (128 modes). Those modes are wide but low-rank compared with n .
- (3) **The information level.** Record deviations from the closure's reference pattern. The deviation set has $0.5n$ elements at layer 1 but only $0.085n$ at layer 16, and the closure predicts its size. Active cardinality is pinned at $n/2 \pm 8$ and carries almost no information.
- (4) **Depth.** Gate dependence persists across at least four layers, and the depth gain of the last four layers is 0.6–0.9. The occupation recurrence cannot truncate late-layer return paths. In the companion's language, m must grow where the hidden operator's gap is small; that happens in the last layers, not the first.

- (5) **What the first pass gives.** The closure is the mean-field wall sum (Proposition 2.2), and its wall measures are accurate in aggregate (Rice–Gauss to 0.1–0.6%). Its error is in the downstream-jump channel, the conditional expectation of the gated product at a wall. That is the quantity a represented operator must improve.

APPENDIX A. INDEPENDENT VERIFICATION OF THE COMPANION’S FINITE CLAIMS

theory24/check_companion.py; exact rationals where the claim is finite.

claim	check	result
eq. (7) two-row heat bath	exact spectrum, $\varepsilon = \frac{1}{10}, \frac{1}{4}, \frac{1}{2}$	pass
eq. (13) five-sector example	jumps $(-1, \sqrt{2}, 0, 0, 1)$, $\mu = 1/(2\sqrt{\pi})$ to 10^{-10} ; Monte Carlo	pass
eq. (11) $\sqrt{t/\pi}$ constant	half-plane, $t = 10^{-2}, 2.5 \cdot 10^{-3}$: ratio 0.998	pass
eq. (16) finite-noise readout	3-layer net, $\rho = 0.6$: 4.6×10^{-3} relative	pass
eq. (17) one-ReLU moment	$\rho = 0.5, 0.9$	pass
eq. (22) posterior gap	three random M , 8 cells: gaps $0.83, 0.65, 0.42 \geq 0.54, 0.26, 0.18$	pass
eq. (28) cycle gap	$m = 6, 12, 40$	pass
Thms 7.1–7.2, eq. (48), table	exact rationals, $m = 0.7$; $\frac{1}{2}$ without hidden reward, $\frac{3}{2}$ without clock	pass
Thms 8.1–8.2, (39), (43), (44)	200 random reversible chains	pass
eqs. (3)–(4)	commutator; $D_{r,s}(c)$ counts	pass

APPENDIX B. TABLES

TABLE 1. Levels and walls per layer, width 1024 ($N = 2^{16}$ inputs and rotation partners per net). Net 0 values; net 0 only (the multi-net run was stopped).

ℓ	r/n	$sd(r)/n$	excitation/ n	R^2 next level: signs / values	wall rate	rate sd/mean	Rice–Gauss / measured
1	0.500	0.0154	0.500 [0.500–0.500]	0.34 / 0.60	0.317 [0.317–0.318]	0.07	1.003 [1.000–1.003]
2	0.504	0.0143	0.314 [0.300–0.317]	0.31 / 0.55	0.322 [0.312–0.323]	0.23	0.999 [0.999–1.001]
3	0.503	0.0127	0.252 [0.248–0.257]	0.32 / 0.56	0.319 [0.314–0.323]	0.38	1.002 [1.001–1.002]
4	0.497	0.0123	0.225 [0.221–0.225]	0.36 / 0.58	0.325 [0.320–0.325]	0.49	1.001 [0.997–1.001]
5	0.498	0.0118	0.191 [0.185–0.200]	0.32 / 0.53	0.317 [0.316–0.330]	0.61	1.005 [0.996–1.005]
6	0.502	0.0109	0.176 [0.173–0.176]	0.31 / 0.51	0.327 [0.320–0.327]	0.67	1.001 [1.000–1.003]
7	0.507	0.0104	0.168 [0.147–0.168]	0.34 / 0.53	0.338 [0.301–0.338]	0.73	1.002 [0.997–1.005]
8	0.464	0.0102	0.157 [0.144–0.157]	0.29 / 0.48	0.345 [0.312–0.345]	0.79	0.999 [0.996–1.004]
9	0.497	0.0092	0.133 [0.126–0.134]	0.28 / 0.48	0.329 [0.299–0.329]	0.89	1.001 [0.999–1.002]
10	0.528	0.0090	0.129 [0.117–0.129]	0.32 / 0.52	0.338 [0.296–0.338]	0.92	1.003 [1.001–1.003]
11	0.518	0.0094	0.127 [0.103–0.127]	0.36 / 0.54	0.355 [0.277–0.355]	0.93	1.003 [0.997–1.003]
12	0.493	0.0090	0.116 [0.103–0.116]	0.30 / 0.50	0.339 [0.288–0.339]	1.03	1.007 [1.002–1.007]
13	0.516	0.0083	0.105 [0.086–0.105]	0.38 / 0.58	0.331 [0.256–0.331]	1.11	1.002 [1.002–1.005]
14	0.522	0.0087	0.094 [0.086–0.096]	0.26 / 0.44	0.312 [0.269–0.312]	1.22	0.999 [0.999–1.007]
15	0.500	0.0075	0.096 [0.088–0.096]	0.33 / 0.52	0.329 [0.284–0.329]	1.20	1.004 [1.004–1.004]
16	0.511	0.0076	0.085 [0.085–0.090]	–	0.309 [0.297–0.309]	1.31	1.006 [1.006–1.008]

TABLE 2. Dependence of the gate law, single-layer wall-chain bounds (per unit rotation angle) and depth gains, net 0.

ℓ	η	η , 16 / 128 PCs out	PR(Corr g)	λ_{lin}	λ_{lin} , 16 / 128 PCs out	level relax.	$\ \text{Corr}(g_\ell, g_{\ell+1})\ $	$\prod_{k>\ell} g_k$
1	1.96 [1.9–2.0]	1.77 / 1.53	720	0.213 [0.210–0.214]	0.23 / 0.33	0.66	2.36	0.030
2	2.65 [2.6–2.7]	2.31 / 1.62	692	0.216 [0.213–0.219]	0.25 / 0.41	0.80	2.97	0.050
3	3.65 [3.6–3.8]	3.02 / 1.66	650	0.201 [0.198–0.205]	0.25 / 0.51	0.93	3.85	0.075
4	4.57 [4.5–4.7]	3.53 / 1.60	622	0.192 [0.187–0.195]	0.25 / 0.63	1.07	4.62	0.109
5	5.14 [4.9–5.5]	3.85 / 1.44	612	0.197 [0.186–0.202]	0.27 / 0.60	1.21	5.29	0.145
6	5.98 [5.7–6.2]	4.08 / 1.35	600	0.190 [0.185–0.200]	0.29 / 0.68	1.31	5.77	0.191
7	6.01 [5.9–6.2]	4.32 / 1.31	596	0.206 [0.204–0.208]	0.31 / 0.72	1.48	6.03	0.240
8	6.79 [6.5–7.0]	4.27 / 1.32	577	0.202 [0.198–0.205]	0.34 / 0.63	1.60	6.54	0.309
9	6.63 [6.4–6.8]	4.22 / 1.37	582	0.224 [0.219–0.230]	0.37 / 0.82	1.78	6.69	0.372
10	7.03 [6.7–7.3]	4.27 / 1.37	567	0.227 [0.219–0.232]	0.40 / 0.96	1.83	7.16	0.446
11	7.84 [7.4–8.3]	4.25 / 1.41	545	0.219 [0.207–0.226]	0.43 / 0.81	1.88	7.65	0.525
12	7.76 [7.4–8.2]	4.23 / 1.32	534	0.232 [0.218–0.241]	0.46 / 0.89	1.99	7.28	0.607
13	7.13 [6.9–7.5]	3.97 / 1.28	520	0.265 [0.249–0.278]	0.52 / 1.16	2.18	6.89	0.685
14	7.17 [6.6–8.0]	4.03 / 1.27	517	0.277 [0.246–0.315]	0.54 / 1.15	2.31	7.15	0.780
15	7.54 [7.1–8.0]	4.08 / 1.25	492	0.276 [0.261–0.284]	0.56 / 1.02	2.53	7.09	0.896
16	7.16 [6.8–7.7]	3.90 / 1.25	471	0.305 [0.273–0.334]	0.61 / 1.13	2.53	–	1.000

TABLE 3. The terminal ellipsoid: visible spectrum of $S = \text{Cov } F$ and the share of first-pass errors in the eigenvector bands $[0, 16), [16, 64), [64, 256), [256, 512), [512, 1024)$; R = Rayleigh ratio $e^T S e / (|e|^2 \bar{\lambda})$.

net	PR	trace, top 16/64/256	bottom half	closure: bands; R	CC1: bands; R
0	48.9	0.418/0.756/0.949	0.0099	0.689 0.149 0.109 0.037 0.015; 68.4	0.209 0.372 0.284 0.104 0.032; 8.3
1	46.2	0.436/0.766/0.950	0.0102	0.743 0.085 0.120 0.040 0.013; 72.9	0.211 0.306 0.337 0.111 0.036; 8.7
2	50.0	0.440/0.772/0.950	0.0104	0.789 0.082 0.082 0.034 0.012; 70.1	0.166 0.356 0.312 0.127 0.039; 8.2
3	51.5	0.425/0.755/0.946	0.0117	0.671 0.123 0.121 0.065 0.021; 58.7	0.248 0.305 0.272 0.132 0.043; 9.7
4	63.7	0.397/0.735/0.937	0.0148	0.622 0.152 0.150 0.054 0.021; 41.1	0.278 0.333 0.254 0.099 0.035; 9.9
5	44.4	0.450/0.773/0.953	0.0083	0.733 0.147 0.075 0.035 0.011; 74.5	0.341 0.310 0.209 0.112 0.027; 11.7
6	52.5	0.422/0.757/0.946	0.0109	0.619 0.132 0.169 0.060 0.019; 51.7	0.318 0.243 0.300 0.108 0.031; 17.6
7	47.6	0.436/0.767/0.953	0.0077	0.771 0.096 0.093 0.032 0.009; 77.6	0.127 0.328 0.387 0.128 0.030; 6.6